

(8.24—8.30)

【8.24| The Guardian】Incredible ways insects are communicating and listening all around us

【Para. 1】

All around us, every day, hundreds of conversations are taking place in languages we do not understand. Birdsong is one you would probably recognise, and you may have heard of honeybee waggle dances. But what about the spider tuning their web to listen for prey, or a waft of arousing perfumes emitted by a passing beetle?

【Para. 2】

To delve into the world of invertebrate communication about food gathering, nest building, defending a territory, attracting a mate, and caring for their young is to realise just how smart and sophisticated they are.

【Para. 3】

Spiders, for example, can “hear ” vibrations on the silken strands of their web using sensors on their legs. With their legs and body weight, the spiders will adjust the silk strings, as we would a guitar string to change its pitch, pulling, loosening or reconnecting strands to adjust how it sounds. In fine-tuning the silk they can know just who or what is on it.

【Para. 4】

Many invertebrates use pheromones – complex ranges of scents – to communicate. Moths can mix a cocktail of perfumes to attract a mate while ants lay chemical trails to guide their sisters to food sources. Some burying beetles share parenting and will communicate using chemical scents, as well as touching.

【Para. 5】

Mimicry is another tool used to deceive. The hoverfly that mimics the yellow and black stripes of the wasp, to deter predators such as birds from eating them, is well known. Spiders and moths can mimic ants in shape, smell and movement so they are able to live and eat in an ant nest rent-free. Several insects mimic parts of a tree to avoid being eaten. Others, such as some praying mantids, can appear perfectly like the

flowers they are waiting in, all the better to lie undetected and pounce on unsuspecting prey.

【Para. 6】

One butterfly caterpillar uses an incredible two-tone “song ” to live undetected in an ant nest. Having tricked the ants, by mimicking ant baby smells, into carrying them into the nest; once installed, the caterpillar will make noises so similar to that of an ant larvae that they get fed and protected by unsuspecting ant workers.

【Para. 7】

And it's not just invertebrate talking to invertebrate. Trees, for example, communicate with bees, using colour and ultraviolet light to attract and guide them to pollinate flowers.

【Para. 8】

Some plants beckon bugs, such as crab spiders, to attack other bugs which are attacking them, while the relationship between the fig tree and the fig wasp is particularly intimate. Every type of fig tree has a unique perfume to call its own wasp species. The tiny female wasp crawls through a hole in the fig flower, often having to lose her wings to get

through, and brings pollen from another tree with her. A whole wasp life cycle will then occur inside what you will eventually eat as a juicy fig.

【Para. 9】

Invertebrates have evolved these relationships over millions of years, finding ways to inhabit their niche better in collaboration with each other and the rest of the world. And it is not all about cooperation. Much communication – from pugilistic ants to worms biting each other – is about status, repelling predators, winning mates and protecting territory, babies or food.

【Para. 10】

Scientists are still discovering the incredible detail of insect communication, without which chaos would ensue. Sadly, when humans interrupt these conversations, whether by interfering with their habitat, using chemicals or causing climate emergency, we cause serious problems – for ourselves as well as the wildlife.

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【8.25| The Guardian】Appy talk: what next for dating as swipe fatigue grows?

【Para. 1】

Bumble was supposed to be different. When the dating app launched in 2014, it offered women a simple proposition: if you matched with a man, you had to make the first move. The rule was meant to make online dating safer and less intimidating, while giving women control over who got to speak first.

【Para. 2】

This week, Bumble abandoned it. The change follows the introduction of Opening Moves, which had already softened the original rule, and comes as Bumble prepares to abandon another defining feature: the swipe. The company said it was responding to changing user

behaviour and wanted to encourage more meaningful connections. But there is another way of reading the changes.

【Para. 3】

Bumble was created to fix what many saw as Tinder's chaotic and aggressive hookup culture by shifting the power dynamic entirely to women. Now it is dismantling those features. And it is not alone. Across the dating app industry, the swipe-and-match model that transformed how people met is being quietly reworked while companies race to put artificial intelligence at the centre of whatever comes next.

【Para. 4】

"Bumble is in salvage mode," said Luke Brunning, a philosopher who studies the ethics of technology and relationships. "They've already reorganised, restructured and cut jobs in response to declining revenue. This risky, standout pivot suggests they're wondering how, despite all that, to keep going."

【Para. 5】

Bumble's paying users fell sharply over the past year and its share price has been under sustained pressure. The company cut about 30% of

its workforce this year. “It’s going to be fascinating to see whether they can weather it,” Brunning said. “But they’re not alone: a lot of the major players are also in salvage mode, trying to figure out how to position themselves in a market that is increasingly fed up with them.”

【Para. 6】

Brunning’s research shows dating apps are now often used grudgingly, because people feel they are necessary, rather than with the excitement that accompanied their rise.

【Para. 7】

The online dating industry – estimated to be worth almost £10 bn globally – is keenly aware of this. Its solution? To go full AI. Grindr – seen as a testbed for the wider industry – is currently transitioning into an “AI- first” platform. Tinder is testing Chemistry, an AI-powered system intended to reduce “swipe fatigue”. Bumble has been developing an AI matchmaking assistant called Bee.

【Para. 8】

Other start-ups are dispensing with swiping altogether. The London-based Fate describes itself as an “agentic AI dating app”: instead

of browsing profiles, users are interviewed by an AI about what they want from relationships to find similarities with potential partners.

【Para. 9】

So why aren't we talking about being on the brink of a brave new world of dating apps, where AI will successfully sift the candidates to find our one true love? Why instead, is the debate about whether the apps will even survive?

【Para. 10】

Adam Peacock, a researcher of technology and intimacy, thinks the issue is that the industry is misreading what people are objecting to. "There's a kind of emerging cultural backlash against allowing AI into our intimate lives as well as additional concern that someone you're talking to on these apps is actually filtering the whole conversation through an AI concierge," he said.

【Para. 11】

Peacock said gen Z in particular had concerns that dating apps were not even being designed to make good connections between users but were, instead, opaquely tweaked to monetise the time users spent on the

sites. That, in a cynical twist, the apps were deliberately profiting from keeping people lonely. “The dating industry has turned intimacy into an enormous source of data,” said Peacock. “Dating apps can know where people live, what they look like, who they are attracted to, what they say to potential partners and, increasingly, how they behave.”

【Para. 12】

But the first generation of dating apps – the era of the profile, the swipe, the match and the message – may be reaching its end. Peacock thinks the apps may evolve into something broader: “more of a social media, lifestyle product than a dedicated dating app”. But the question is whether any iteration of dating app will make dating better, or whether the extra layers of AI will simply function as a temporary facade on an exhausted model.

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【8.26| The Guardian】Will AI destroy film industry or usher in a golden age?

【Para. 1】

Near the colonnaded entrance to Sony Pictures' Culver City film studio, a tech-powered rival has sprung up this summer. The artists at Promise, a new studio feeding off the potential of AI, can see the lots where Singin' in the Rain and Men in Black were shot, but no longer do they need to get beyond the studio's high walls to make their visions come true.

【Para. 2】

From far more modest premises, they are making movies with rapidly accelerating AI models that create backgrounds, special effects and even "synthetic performers" – using US models but also driven by market-leading Chinese AI. Backed by Google, Silicon Valley venture

capitalists and Disney, Promise is one of a rising number of AI film studios in Hollywood.

【Para. 3】

It is a controversial trend. Leading directors including Christopher Nolan and Guillermo del Toro have criticised generative technology, actors and writers are fearing for their jobs, and the Democrat congresswoman Laura Friedman, who represents Hollywood's actors, writers and directors, warned: "We can't wait until we see tens of thousands of American workers displaced."

【Para. 4】

But the excitement and energy about AI was palpable this week at Promise's new base, when a human actor, Tori Thomas, waited for her call amid rows of computers on the studio's AI-enabled horror film, Touch Grass, with a budget in the low millions of dollars.

【Para. 5】

The sound stage was a corner of the office hung with white sheets. A small camera followed Thomas as she scrambled in a chase scene, then the AI superimposed her movement in real-time on to a field of swaying

grass generated by a new Chinese AI model, Seedance 2.5, which looks as if it will slash production times. Whole scenes – actor and backdrop – were visible on the director’s handheld monitor as if they were filming in real locations. With a glance at the screen, Thomas could immediately see the backdrop she was acting with.

【Para. 6】

Promise is not alone in embracing the technology, which Netflix last week said it had used in 300 of its 1,000 titles in 2026. Imagine, the 41-year-old studio run by Brian Grazer and Ron Howard, the director of *The Da Vinci Code* and *A Beautiful Mind*, now has an AI studio, Obsidian, operating from space in its Beverly Hills HQ. Imagine has partnered with Obsidian to make *The Code*, an AI-enabled, Howard-directed animated feature based on drawings made in Vietnamese captivity by US prisoners of war.

【Para. 7】

Brad Pitt this week said that, although there was a lot of resistance, “AI is going to be the very thing that, if used as a tool, is going to help these mid-budget films, \$40m to \$90m-budgeted films, get made.” That chimed with predictions from a new wave of AI native film-makers, such

as Kavan “the Kid” Cardoza, 37, who has set up the Phantom X studio. He predicted a creative renaissance circumventing the studio giants not seen in more than 30 years.

【Para. 8】

The Promise shoot involved veteran film-makers including Joel Hynek, an Oscar-winner who worked on the manual special effects for Arnold Schwarzenegger’s 1987 action-horror film Predator. “When we went from optical to digital film, we thought we’d gone to heaven,” he told the Guardian.

【Para. 9】

Others, such as Nolan, voice concern. “I’ve never seen a technology advancing so rapidly, so completely rejected by the public,” the director of The Odyssey has said. “Everybody’s suspicion of it is so extreme, particularly young people.”

【Para. 10】

There has been opposition to an AI resurrection of the late actor Val Kilmer to star in a new feature and the emergence of the AI “actor” Tilly

Norwood, with the actor Emily Blunt pleading: “Please stop taking away our human connection.”

【Para. 11】

But the new AI studios are undeterred. Dave Clark, director of Promise’s Touch Grass said that with “great dialogue and backstory” AI-generated characters can become interesting and believable. The actors’ union SAG-Aftra will no doubt beg to differ, but Cardoza predicted “next year [AI actors will] be just as good as real actors”. George Strompolos, co-founder and chief executive of Promise, estimated hybrid AI films – human actors alongside AI technology – will prove 20% to 50% cheaper than conventional production. That could free film-makers from requiring major studio investment.

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【8.27| Economist】Frozen out

Japan's Gen X workers are struggling

【Para.1】

For nearly two decades Torigoe Atsushi, who is 50, has worked in the stockrooms of a fashion firm in Tokyo. His income rose only modestly, and it recently fell after his employer cut bonuses and reworked its pay scheme; he now earns about ¥3.6m (\$22,600) a year. Then he spotted the firm's recruitment ads, offering newcomers higher salaries. "It's great that younger people are getting better pay," he says. "But why not raise mine?"

【Para.2】

Mr Torigoe belongs to Japan's "ice-age" generation: workers who entered the labour market between 1993 and 2004, after the country's economic bubble burst and many companies stopped hiring. The

ice-agers often endured years of insecure work and low pay. Now in their 40s and 50s (thus Generation X), they are losing out again. Japan is emerging from decades of deflation, prices are rising and companies are handing out their biggest pay increases in a generation. But the ice-agers are not benefiting. Worse may come when this cohort of baby-boomers' children—the last bulge in Japan's population pyramid—drifts into retirement.

【Para.3】

The ice-agers' fortunes contrast starkly with those of the generation that came just before. During the bubble years of the 1980s, companies hired voraciously. College students found their mailboxes stuffed with offer letters, or were treated to fancy meals by employers keen to woo them. Everything changed as the bubble popped in the early 1990s. Japan's rigid employment system made it hard for firms to shed existing workers, so they cut hiring instead. The ratio of job openings to college applicants fell from nearly three at its peak to below one by 2000. Nagai Toshihisa, who graduated in 1999, shortly after the Asian financial crisis, abandoned his hope of working for a securities firm. "It felt utterly stupid looking for jobs," he recalls. Many in his generation sent out dozens, even hundreds, of applications, to no avail. It did not help that the ice-age cohort was such a big one, adding to the oversupply.

【Para.4】

Many graduates who failed to secure regular jobs ended up in temporary or contract work. Japan's labour market sharply distinguishes between regular workers, who enjoy strong legal protections and seniority-based pay rises, and irregular ones, who often do similar work for much less money and are the first to be laid off in downturns. Such jobs often used to be filled by housewives supplementing their husbands' salaries. Increasingly, they were taken by young people who needed to support themselves, says Kondo Ayako of the University of Tokyo. The term "freeters"—youngsters drifting between casual jobs—became popular, as did "parasite singles", who lived off their parents well into adulthood. There was also growing concern about hikikomori, social recluses.

【Para.5】

Time and acute labour shortages have repaired some of the damage. The share of ice-agers in regular employment has largely caught up with that of older cohorts, says Shimoda Yusuke of the Japan Research Institute. But their earnings have not. Those who spent long stretches in irregular work tend not to fully catch up even after switching to the regular track. Those in regular jobs faced another problem: the upper

ranks of their companies were crowded with the bubble-era workers, delaying promotions for those who followed. Japan's low labour mobility has meant the unlucky cohort cannot easily seek better pay elsewhere.

【Para.6】

Japan's economy is changing. After three decades of stagnation, wages are rising as inflation returns, and job-hopping is becoming more common. Yet these improvements are mostly benefiting the young. Companies short of labour are competing fiercely for fresh blood, and paying more to attract and retain it. Middle-aged employees are much less likely to switch jobs, giving firms little incentive to raise their wages. According to Kumano Hideo, an economist at Dai-ichi Life Research Institute, nominal wages for college-educated workers in their 20s and 30s were 10-16% higher in 2025 than five years earlier. Meanwhile, those in their early 50s suffered a nominal decline of 1.3%. Abe Katsuya, 48, says the large pay rises dished out to the young give him "mixed feelings". He is part of the "Mammoth Club", a support group for ice-agers who identify with the woolly beasts that endured harsh times.

【Para.7】

All this has left the ice-agers with low lifetime earnings (and therefore little in the way of savings). That points to a looming crisis as

they reach old age. Japan's pension system ties payouts to lifetime earnings, so decades of suppressed pay will translate into low pensions. "Those who happened to come of age in a downturn are likely to carry that handicap for the rest of their lives," laments Mr Kumano. Getting them to work longer, as a growing number of old people already do, is one fix but it will not solve everything. Ms Kondo warns that many may end up on welfare, which is not designed to support the elderly over extended periods. That would be costly; for instance, the state pays welfare recipients' medical bills in full. And because the generations behind the ice-agers are smaller, fewer workers will have to support more pensioners.

【Para.8】

The woes of the ice-age generation have long had the government's attention. For decades, authorities have come up with programmes to help them find jobs and nudge reclusive ones to connect with the outside world. Yet as the ice-agers grow older they face new problems. Beyond thin pensions and savings, questions loom, for instance, over housing; compared with older Japanese, this cohort is less likely to own homes, and Japan's rental market is unfriendly towards older tenants. The government is belatedly adjusting: in April, it produced a three-year plan for the ice-agers, spanning measures from asset-building to housing. But

a lot more serious thinking will be needed.

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【8.28| Economist】Dinosaur gastroliths

Rock-solid evidence for the origins of birds

【Para.1】

How do you know what a dinosaur ate? Guts rot away after death, so the details of dinosaur digestion remain obscure. The best indicators are teeth. Sharp, pointed teeth probably ripped chunks of flesh off prey for swallowing whole. Robust grinding surfaces suggest lots of chewing was involved. Yet many dinosaurs belonging to a group called the theropods, which gave rise to birds, lost their teeth altogether and evolved beaks.

This has made it hard to work out what they ate and how they digested it.

【Para.2】

There is, however, a second clue. Lots of dinosaurs had stones, known as gastroliths, in their stomachs. These are assumed to have assisted with digestion, and thus have their own tales to tell. And modern birds and crocodilians (which are both, like dinosaurs, members of a larger evolutionary group called the archosaurs) often have gastroliths, too.

【Para.3】

Takasaki Ryuji at Okayama University of Science, in Japan, decided to investigate. In a study published in *Paleobiology*, he and his colleagues examined the gastroliths of 46 modern archosaur species (42 birds and four crocodilians) and compared them with those from 16 species of dinosaur.

【Para.4】

The birds included herbivores, omnivores and carnivores. The crocodilians were all carnivorous. The shapes of the stones varied. As might be expected, if an animal had a highly muscular stomach (ducks and geese), its gastroliths tended to have been worn into the sorts of

rounded shapes displayed by beach pebbles. Weaker stomachs (seabirds and crocodiles) contained more angular stones.

【Para.5】

Looking at the dinosaurs, Dr Takasaki and his colleagues found that ornithischians (think Triceratops), a group which their teeth suggest were herbivores, retained angular gastroliths. These creatures, this implies, relied mostly on their jaws to grind up their food, leaving their stomachs to apply merely the finishing touches.

【Para.6】

Long-necked sauropods (think Diplodocus) also had angular gastroliths. That is more of a puzzle, for these animals had pencil-like teeth using which, it is assumed, they stripped the leaves off tree branches that less cervically elongated animals were unable to reach. The explanation, Dr Takasaki suspects, is that like those of modern ungulates, sauropods' stomachs depended mainly on fermentation to break food down.

【Para.7】

Stones in the guts of theropods tell yet another story. Tarbosaurus, a hypercarnivore similar to Tyrannosaurus, and similarly endowed with

sharp, pointed teeth, had angular gastroliths. However, theropods with beaks (of which science recognises at least four groups) had rounded ones. Indeed, Dr Takasaki saw a clear relation between tooth loss and stone roundness, as the work of grinding up food was transferred from jaws to stomach.

【Para.8】

That shift may have helped make possible the evolutionary success of birds. Teeth are heavy, as are the jaw muscles those teeth require to do their job. This puts a lot of weight near an animal's front. Shifting the job of grinding, and the muscles and hard surfaces involved, to the stomach centralises an animal's centre of gravity in a way that would make it easier for it to rise from the ground.

【Para.9】

Curiously, it is clear from the fossil record that the very earliest birds of all did not have beaks. But the fact that so many of their theropod relatives did, and that birds themselves evolved them at least twice, suggests the transition was easy. It may thus be that part of the secret to taking successfully to the skies was, quite literally, having the stomach for it.

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【8.29| Economist】On shifting oil sands

Canada's oil industry is booming. Can it last?

【Para.1】

“Please god, give me one more oil boom—and I promise not to piss it all away next time” reads a popular bumper sticker in Canada's oil patch. Recent results from the country's oil producers show that plea has been answered. Oil prices have jumped as the passage of tankers through the Strait of Hormuz has been choked off. Last month Cenovus Energy reported that operating profits in its most recent quarter had nearly tripled compared with a year before. Canadian Natural Resources posted its

best-ever quarterly adjusted operating profit. Moreover, the easing of political headwinds in Canada may improve the chances of the boom lasting.

【Para.2】

Canada has the world's fourth-largest proven reserves, with 97% found in the oil sands in a remote corner of the land-locked province of Alberta. Here the bitumen, mixed with sand and clay, is almost solid, and is mined rather than tapped from the ground. It is costly and dirty to extract and process. Despite the drawbacks, high prices combined with a weakening of the Canadian dollar mean that "it hasn't been this good to be an oil-sands producer in a very long time," says Andrew Leach of the University of Alberta.

【Para.3】

Getting more oil to market is also set to become easier. Justin Trudeau, Canada's former prime minister, was tepid in his support for new pipelines, fearing the consequences for the climate. In 2018 regional political opposition led Kinder Morgan, an American firm, to suspend work on a conduit from Alberta to the west coast in British Columbia. That forced the national government to step in and complete it. Oil eventually started flowing in 2024.

【Para.4】

Lately the political mood has shifted. A trade war with America, the destination for 90% of Canadian oil, has shifted the federal government's focus to reducing dependence on its southern neighbour. That includes support for building new infrastructure to transport oil (and gas) to ports on Canada's west coast, from where it can be shipped to Asia. Mark Carney, Canada's current prime minister, is intent on slashing the red tape that holds up projects, including streamlining environmental permitting and consultations with indigenous communities. The speedier process will apply to a new pipeline under development to British Columbia, announced last month. Lisa Baiton of the Canadian Association of Petroleum Producers, an industry group, has spoken of a "generational opportunity" to rally political and public support behind new oil projects.

【Para.5】

The promise of a federal government more friendly to pipes may not be enough, however. Enbridge, an energy- infrastructure firm, has paused construction of a pipeline to the east, citing a lack of commitments by producers to provide enough oil to make it worthwhile.

【Para.6】

The unique challenges of the oil sands provide an explanation. Investment in new sites has been virtually non-existent since the oil price crashed in 2014 as demand weakened while America's shale oil gushed and OPEC opened the taps. "It cannot be overstated how much that changed the industry," says Kent Fellows of the University of Calgary. Moreover, projects in the oil sands take far longer to develop than conventional wells. They require enormous amounts of capital that could be tied up for five to ten years before profits flow. Uncertainty over the future price of oil—and the danger that a change of government might lead to the reinstatement of tighter regulations—make for risky bets.

【Para.7】

Nevertheless, at the Calgary Stampede, an annual rodeo festival held in July that serves as the industry's unofficial barometer, parties hosted by oil companies were back to their riotous best after quietening down in recent years. Their sponsorship of rowdy chuckwagon races broke records. That is as good a sign as any that another boom is in full swing.

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【8.30| Economist】Without a trace

Fewer contrails would be good for the climate

【Para.1】

Good news on global warming is thin on the ground. But there may be some up in the air. Operation Blue Skies, announced on August 18th, is a scientific study being undertaken by Britain's air-traffic-control service, NATS, and various organisations where people study the effects on climate of the streaks of cloud created by aircraft—known as contrails.

【Para.2】

Over the next two winters the consortium aims to see if it is possible to reduce the number of persistent contrails—the class with the largest such effects—over a swathe of the North Atlantic, thereby measurably reducing the climate impact of civil aviation. If this experiment works, a worldwide application of the same techniques might cut the climate

effects of air travel by as much as half.

【Para.3】

On a sunny day the wispy cirrus clouds into which some contrails are transformed can look as if they might be cooling the planet in the manner of a delicate parasol. And some contrails do indeed cool things down beneath them a bit. But they also warm things up by trapping some of the upwelling infrared radiation that would otherwise escape into space. On average, that warming effect, found in all contrails (and the cirrus clouds they turn into), handsomely outweighs the cooling provided by some. In the climate scheme of things, contrails are a liability.

【Para.4】

Happily, it appears that much harm can be avoided by small changes to flight paths. Persistent contrails appear only in what are known as “ice-supersaturated regions” (ISSRs). Often planes can avoid such regions just by flying a little bit lower. The purpose of Operation Blue Skies is to find out how well the presence of ISSRs can be predicted, how effective such little shifts in altitude can be in avoiding persistent contrails, and how easily and safely the changes in flight path needed to bring them about can be integrated into routines for airlines and air-traffic control.

【Para.5】

The cut of up to half in the climate-effects of air travel would come at a very small cost. The number of flights at lowered altitude might be small—as little as 5% of the total. Because altitude buys efficiency, those flights would consume more jet fuel, but less than 2% more.

【Para.6】

Burning even that little extra fuel would be anathema to airlines. So some suggest the costs of action might be minimised, and the airlines incentivised, by setting up a market which somehow links contrail cooling to carbon prices. The European Union, which has had some success and plenty of experience with carbon markets, recently took a small step towards something along these lines. It is also the sort of idea to which economists—and *The Economist*—tend to be partial. But it would be a mistake nonetheless.

【Para.7】

Markets for pollutants have worked best when both the aim and the remedy are fairly simple—most famously, reducing sulphur-dioxide emissions from power stations in America. Contrails, by contrast, are complicated. Though there is no doubt that their net effect is warming,

saying how much warming a particular contrail might produce is often difficult and sometimes next to impossible.

【Para.8】

The difficulties are made yet worse when the contrail being assessed is a purely notional one—as would be the case for the contrails avoided by changing flight paths. Valuing counterfactuals is always hard.

【Para.9】

If contrail-avoidance turns out to be as useful as it looks, it would be much simpler to simply have air-traffic controllers assign routes that limit persistent contrails when it is possible and safe to do so. If some airlines choose to avoid the affected routes, others will surely be happy to take their place. And if such a scheme were also to show the benefits of innovations in air-traffic control (quite a few have been suggested) the industry would end up better off, as well as less polluting.

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